

Detailed and Updated Summary:
Autonomous view planning methods for 3D scanning
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1 Abstract and Introduction

“Recent advancements in three-dimensional (3D) scanning technology have broadened its applicability across fields like reverse engineering and autonomous robot navigation, which require a 3D representation of real-world objects. [...] This paper reviews autonomous view planning methods for 3D scanning of unknown geometries.”

1.1 Objective of the Paper

- **Systematic review of autonomous view planning methods**
- Focus on **unknown objects** (unknown geometries)
- Applications: **construction, digital twin (1:1 real-time reconstruction), inspection**
- Provided tools: **general pipeline, taxonomy, Sankey diagram, future directions**

2 Review Methodology

All existing papers were studied and filtered to conduct the study, dividing the process into 4 phases:

Fig. 1: Article selection flow

- **Identification from:**
Web of Science → **n = 170**
Scopus, Google Scholar, IEEE, Elsevier, Springer → **n = 136**
TOTAL → **n = 306**
- **Screening:**
Duplicates removed (n = 94) → **n = 212**
Removed studies on general 3D scanning and not on scanning of partial objects (n = 98) → **n = 114**
- **Eligibility:** Excluded 2D, manual, non-3D (n = 29) → **n = 85**
- **Included:** **n = 85** articles for Sankey, **n = 85** total

2.1 Temporal Trend

- **2022 Peak:** >30 articles/year
- Drivers: **UAVs, low-cost sensors, deep learning**

3 Autonomous View Planning Pipeline

The **general pipeline** describes the four fundamental phases that every autonomous view planning system must execute to achieve a complete and efficient 3D scan. Each step is mandatory and corresponds to the sections discussed in the survey.

Fig. 2: General pipeline

1. **3.1 Environment setup:** definition of the environment, the target, the sensor, and occlusions.
2. **3.2 Problem modeling:** choice of representation (voxel grid, octree, mesh).
3. **3.3 Algorithm:** selection of the view planning approach (NBV, model-based, hybrid).
4. **3.4 Termination:** stopping criterion based on reconstruction quality.

3.1 Environment setup

In this phase, the fundamental components are established before performing the scan:

- **Target:** the object or environment to be acquired (small objects, rooms, buildings, urban scenes).
- **Sensor:** type of 3D sensor used (LiDAR, RGB-D, stereo), with parameters such as FOV, depth range, noise.
- **Occlusions:** surfaces, objects, or geometries that block the field of view and affect visibility.

3.2 Problem modeling

The **knowledge representation** that the system will use during scanning is defined:

- **Voxel grid:** discrete volumetric representation (occupied/free/unknown).
- **Octree:** hierarchical voxelization that reduces memory and computation.
- **Mesh:** triangulated surfaces, used in advanced reconstructions or model-based methods.

3.3 Algorithm

The computational phase in which the next optimal view is decided:

- **NBV (Next-Best-View):** selection of the view with maximum expected information (information gain).
- **Model-based:** use of a CAD or partially known model to guide the choice.
- **Hybrid:** combination of multiple strategies, often the most robust choice.

3.4 Termination

The procedure terminates when a target reconstruction quality is achieved:

- sufficient surface coverage;
- entropy reduction below a threshold;
- minimum percentage of unknown voxels;
- error (RMSE) of the reconstructed mesh below the threshold.

4 Taxonomy

The following table classifies the analyzed studies according to four fundamental criteria: scenario scale, target object type, application, and the amount of available initial knowledge.

Table 1: Taxonomy of studies

Criterion	Category	Included	Excluded	Notes
Scope of survey (scenario scale)	Micro scale	✓		Millimetric objects, high precision.
	Jewelry, dentistry	✓		Very complex scans with severe occlusions.
	Indoor scale	✓		Rooms, corridors, indoor environments.
	Urban scale	✓		Cities, buildings, roads, outdoor.
Target object	Small	✓		Small/medium-sized objects.
	Indoor – Building	✓		Architectural structures and environments.
	Urban scale	✓		Large-scale outdoor scenes.
Application	Object reconstruction	✓		Complete object scanning.
	Mapping, navigation	✓		Mobile robotics, exploration.
	Pose estimation	✓		Target alignment and localization.
	Unknown	✓		Cases not clearly defined in the paper.
Knowledge	Known (3D CAD)	✓		Complete CAD model available.
	Partially known	✓		Incomplete or approximate models.
	Unknown	✓		No initial knowledge: pure exploration.

Interpretation of the taxonomy

- The **scope** indicates the scenario scale: from microscopic to city scale.
- The **target object** classifies the type of object to be scanned.
- The **application** indicates the purpose of view planning: reconstruction, mapping, navigation, etc.
- The **knowledge** represents how much information is possessed at the start (from complete CAD to zero knowledge).

5 Historical Evolution

Table 2: Evolution of methods

Year	1979–1989	1990–2002	1988–2010	2019–2023
Papers	88	92	250	149
Knowledge	Known	Known	Unknown	Unknown
Manipulator	Robot arm	Robot arm	Ground robot	UAV
Application	Inspection	Inspection	Reconstruction	Exploration/Mapping

6 Sensors

Table 3: Sensors

Type	Advantages	Limitations
ToF camera	Direct depth	Light-sensitive
Structured light	High precision	Range <5m
Laser triangulation	High accuracy	Slow
Stereo camera	Passive	Requires texture

7 Algorithms

The view planning approaches examined in the survey are grouped into three main classes: *model-based*, *non-model-based (NBV)*, and *hybrid*. Here we provide an operational and detailed description of the techniques, typical computational steps, and relative variants that emerged from the analyzed literature.

7.1 Model-based

The **model-based** method assumes the availability of an a priori 3D model (e.g., CAD) and exploits this model to plan the entire scanning path. The typical operational phases are:

1. simulation of candidate views by sampling the model surface;
2. for each view, estimation of visibility / coverage via ray-casting on the model;
3. choice of a sequence of views that minimizes the number of poses or total time while respecting kinematic constraints;
4. collision-free trajectory planning optimized with respect to the known model.

Advantages: low online complexity (much of the computation is offline), high efficiency when the model is accurate. Limitations: not applicable to unknown or highly deformable targets; sensitive to discrepancies between CAD and reality.

7.2 Non-model-based (Next-Best-View, NBV)

The **NBV** algorithms progressively build a representation (voxel, octree, mesh, or TSDF (Truncated Signed Distance Function)) and decide the next best view based on the expected information. A typical NBV cycle is:

1. initial acquisition and integration into the incremental model (e.g., octomap (a map divided into cubes called voxels, each voxel is a node of a tree, and cubes sharing the same state are the same voxel, saving memory) or TSDF);
2. identification of low-confidence / unknown regions (frontier detection, unknown voxels, low-density surface regions);
3. generation of candidate views (a sphere is built around the estimated center of the object and new views are sampled from there (simple), RRT nodes (considering the robot’s workspace and motion planning reduces wasted poses), frontier points (focuses on areas with fewer voxels, useful for exploration), orientations from normal clustering (groups point cloud points into sections by calculating the average normal, which becomes a surface to observe, very useful for large spaces but inefficient for noisy surfaces));
4. evaluation of views with a *utility function* (examples: unknown-voxel count, unknown surface area, information gain, entropy reduction, expected surface coverage);

5. selection of the NBV (maximum gain normalized by cost) and trajectory planning towards it.

Important technical details:

- **Utility metrics:** *unknown voxel count* (counting unobserved voxels within the frustum), *unknown surface gain* (surface area not yet seen), probabilistic occupancy estimations; many implementations use ray-tracing on OctoMap or TSDF for these calculations.
- **NBV selection with cost:** the selection often optimizes an information/cost ratio (e.g., expected information per unit distance), rather than just raw information.

Advantages: flexibility in the absence of a model; suitable for unstructured scenarios. Disadvantages: high online computational complexity and scalability issues in large scenes.

7.3 Hybrid

The **hybrid** approaches mix a priori knowledge (even approximate) with NBV techniques. Typical pattern:

- use the partial/CAD model to plan the first views and rapidly reduce global uncertainty;
- switch to NBV strategies to complete unexpected or non-compliant regions;
- maintain a utility function that combines gain from model coverage and gain from local observations.

This class is indicated in the paper as the most promising for practical applications that must tolerate discrepancies between the model and reality.

7.4 Auxiliary Techniques and Implementation Variants

Surface prediction / inference Some works infer the continuation of the surface (air-vector (very simple, looking perpendicularly to the normal assuming the surface continues like what we already know), trend surface (1st-2nd degree curves are applied to predict surfaces based on the trend of the known surface), primitive fitting (attempting to fit the model to primitive shapes like cylinders or spheres)) along the edge of the reconstructed region to guide the NBV (e.g., use of OCS, quadratic patches, super-ellipsoids). These predictions reduce the number of necessary views by estimating where the surface will continue.

Machine learning and RL Neural networks are used to:

- predict NBV directly from point clouds or partial representations (e.g., PC-NBV, networks that estimate information gain without ray-casting);
- build topologies (GNG: an unsupervised algorithm that grows nodes and adapts them as it discovers) or view quality evaluators;
- end-to-end learning (from point cloud directly to **NBV** without intermediate steps) in scenarios where simulated/real-world datasets are available.

7.5 Integrated Path Planning

After the discrete choice of views, it is necessary to plan collision-free trajectories. The documented methods include:

- classical graph-based: Dijkstra, A* (on grids or roadmaps).
- sampling-based: RRT, RRT*, PRM, and lazy variants; used both to generate NBV candidates and to plan trajectories.

- potential field / harmonic functions for rapid exploration (with attention to local minima).
- continuous trajectory optimization (differentiable info-gain, elastic band, refinement).
- robust to *motion uncertainty*: strategies that integrate error distributions to prefer trajectories that minimize information loss due to imprecise localization.

8 Challenges and Future Directions

The paper concludes by analyzing practical limitations and proposing priority research lines. Here I report in detail the technical challenges and future directions indicated, with concrete suggestions for implementation.

8.1 Main Technical Challenges

1. **Partially known objects / model-reality discrepancies**: consistently integrating CAD information and data acquired online remains complex; robust fusion methods and criteria to weigh the confidence in the a priori model are needed.
2. **Real-time and hardware limitations**: the need to decide the NBV in real time on mobile robots requires:
 - computationally lightweight utility metrics (e.g., approximations of information gain);
 - candidate pruning strategies and approaches to limit the number of evaluations (e.g., local RRTs, receding horizon).
3. **Scalability and memory**: volumetric representations (voxels) and ray-casting over extended spaces are expensive; hierarchical octrees/TSDFs and surface-based methods are recommended to optimize memory/precision.
4. **Standardized evaluation and metrics**: a common standard to compare methods is lacking (quality vs. efficiency trade-off); benchmarks with shared metrics (coverage, RMSE, time, distance traveled, energy) are needed.
5. **Simulator reliability and the sim2real gap**: many works rely on simulators that are not faithful to sensor characteristics; for data-driven or RL approaches, realistic simulators reflecting noise, reflections, and materials are required.

8.2 Suggested Research Directions

- **Mature CAD+NBV hybrids**: develop pipelines that initially exploit CAD but automatically activate NBV procedures when discrepancies are detected; provide *model trust* modules that update the CAD weight based on consistency with observations.
- **Learning-assisted methods for NBV**: networks that estimate information gain or predict NBV from partial maps can eliminate expensive online ray-casting; however, they require diverse datasets and real-world validation.
- **Planning with uncertainty and perception-aware path planning**: integrate localization assessments into the cost function (to choose trajectories that maximize information while maintaining good localization). Examples: perception-aware RRT*, receding-horizon strategies. (Planning with uncertainty refers to scenarios where an action does not go exactly as we think it will).
- **Evaluation standards and shared datasets**: create benchmarks with indoor/outdoor scenarios, different sensors, and ground truth to compare methods on quality, efficiency, and robustness; this will also facilitate the adoption of data-driven approaches.

- **Better sensory simulators:** invest in simulators that model noise, reflection, transparency, and real artifacts (e.g., BlenSor is a step in this direction) to reduce the sim2real gap.

8.3 Implementation Notes for ROS/MoveIt Applications (Practical Advice)

- use `octomap` or hierarchical TSDF for volumetric representation and to reduce the cost of ray-casting; use octrees for variable memory.
- consider receding-horizon strategies (not for the whole scanning process but only for a limited time horizon) for NBV on MAVs (Micro Aerial Vehicle → small drone flying in indoor/outdoor environments) or ground robots to limit latency.
- integrate localization metrics into the cost (e.g., choose trajectories with low pose uncertainty to guarantee effective ICP (Iterative Closest Point) registrations).
- when adopting ML methods, collect simulated data with high-fidelity virtual sensors and extensively validate in the real world before deployment.

9 References

References

- [1] Lee, I.D., Seo, J.H., Yoo, B.Y. (2024).
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